Higher National Diploma in (Electrical) Engineering

Fourth Year, First Semester Examination – 2018

EE 3211- Electrical Machine Control and Protection.

Instructions for Candidates:
Answer any five questions.
All questions carry equal marks.
Calculators are allowed.

No of pages : 03
No of questions : 06
Time: Three (03) hours

---

1. i) Explain the field control method of DC shunt motors and the reason for not archiving the speeds less than the rated speed of motor.

(5 Marks)

ii) Explain the PWM method of Speed Controlling applying to a DC separately excited motor with a neat block diagram. (5 Marks)

iii) Two DC series motors with different air gaps, but otherwise identical, run at 700 and 750 rpm, respectively, when taking 50A at 500V. The total resistance of each motor is 0.36 ohms. If the motors are mechanically coupled, and connected in series to a 500V supply to take a current of 50A, calculate.

a) The speed at which the set will run.

b) The Voltage across each machine. (10 Marks)

2. i) List five advantages of using Electric Braking over Mechanical Braking.

(5 Marks)

ii) Explain in detail the Plugging braking applied to a DC Separately Excited motor.

(5 Marks)

iii) A 440 V dc Separately Excited motor has a rated armature current of 76 A at a speed of 1000 rev/min. The armature resistance of the motor is 0.377Ω. Assume that the load torque is gravitational. A dynamic braking technique employing a braking resistance of 2 Ω is used. Calculate the new steady state speed.

(10 Marks)

3. i) List the three basic service conditions of Electric Drives and explain each of them with relevant load curves (5 Marks)

ii) A motor has a thermal heating time constant of 45 minutes. When the motor runs continuously on full load, its final temperature rise is 80 C. Calculate,

a) What would be the temperature rise after 1 hour if the motor continuously runs at full load?

b) If the temperature rise on 1 hour rating is 80 C, find the maximum steady state temperature at this rating.

c) How long the motor take for its temperature to rise from 50 to 80 C if it's working at its 1hour rating.

(15 Marks)

4. i) Explain the working of a poly phase induction motor with relevant vector diagrams. (5 Marks)

ii) Explain the rotor resistance speed control method applied in slip ring Induction motors.

(5 Marks)

iii) The rotor of an eight pole 50 Hz, three phase induction motor has a resistance of 0.2 ohms per phase and runs at 720 rpm. If the load torque remains same calculate the additional rotor resistance that will reduce its speed by 10%. State any assumptions in your calculation. (10 Marks)

5. i) What are the factors to consider when synchronizing a Synchronous generator to mains. (5 Marks)

ii) Explain with a neat sketch the dark lamp method of synchronizing a generator to mains. (5 Marks)

iii) Explain the construction of an HRC fuse applied in motor protection. (10 Marks)

6. i) Explain the use of thermal overload relay in a motor protection scheme.

(5 Marks)

ii) Explain the necessity of a Reverse power relay in a generating system.

(5 Marks)

iii) Explain with a neat sketch how the stator temperature thermal protection is achieved in generators.

(10 Marks)